

Predictors of Heart Disease Among Adults:

A Logistic Regression Analysis Using R

A Public Health Data Analysis Report

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Statistical Methods: Descriptive & Bivariate Analysis | Multivariable Logistic Regression

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Chapter 1: Background

Cardiovascular diseases (CVDs) are among the leading causes of death worldwide and represent a major public health challenge. Heart disease contributes substantially to morbidity, mortality, and healthcare costs, making early identification of high-risk individuals essential for prevention and treatment.

Several demographic and clinical factors, including age, sex, blood pressure, cholesterol level, and exercise-related characteristics, have been associated with heart disease. Understanding these relationships can support evidence-based decision-making and improve health outcomes.

Biostatistical methods play an important role in identifying disease predictors and evaluating health risks. This project applies logistic regression analysis to a heart disease dataset to investigate factors associated with heart disease and assess the predictive performance of the model.

Chapter 2: Objectives

General Objective

To identify factors associated with heart disease among adults using logistic regression analysis.

Specific Objectives

- To describe the demographic and clinical characteristics of study participants.
- To assess the association between selected predictor variables and heart disease.
- To identify independent predictors of heart disease using multivariable logistic regression.
- To evaluate the predictive performance of the final model.

Chapter 3: Dataset Description

Data Source

The data used in this project were obtained from a publicly available heart disease dataset on Kaggle. The dataset contains demographic and clinical information commonly used in cardiovascular disease research. It consisted of 918 observations and 11 predictor variables.

Study Population

The dataset includes individuals who underwent cardiovascular assessment. Each observation represents one participant and contains information on heart disease status and potential risk factors.

Variables

The outcome variable was heart disease status (Yes/No). Potential predictor variables included age, sex, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, resting ECG results, maximum heart rate, exercise-induced angina, oldpeak, and ST slope.

Table 1. Description of Variables Included in the Analysis

Variable	Description
Heart Disease Status	Outcome variable (Yes / No)
Age	Age of participant in years
Sex	Biological sex of participant
Chest Pain Type	Category of chest pain experienced
Resting Blood Pressure	Blood pressure measured at rest (mm Hg)
Cholesterol Level	Serum cholesterol level (mg/dl)
Fasting Blood Sugar	Fasting blood sugar > 120 mg/dl (Yes/No)
Resting ECG Results	Resting electrocardiographic results
Maximum Heart Rate	Maximum heart rate achieved during exercise
Exercise-Induced Angina	Presence of angina induced by exercise (Yes/No)
Oldpeak	ST depression induced by exercise relative to rest
ST Slope	Slope of the peak exercise ST segment

Chapter 4: Methods

Study Design

This project employed a cross-sectional analytical approach using a publicly available heart disease dataset.

Data Quality Assessment

The dataset was examined for completeness, missing values, duplicate observations, and consistency before analysis.

Statistical Analysis

Descriptive statistics were used to summarize participant characteristics.

Bivariate analyses were conducted using independent sample t-tests and chi-square tests.

Multivariable logistic regression was performed to identify independent predictors of heart disease.

Model performance was evaluated using accuracy, sensitivity, specificity, ROC curve analysis, AUC, and Variance Inflation Factors (VIFs).

Chapter 5: Results

5.1 Descriptive Characteristics of Study Participants

A total of 918 participants were included in the analysis, of whom 508 (55.3%) had a confirmed diagnosis of heart disease and 410 (44.7%) did not. Table 2 summarizes the baseline demographic and clinical characteristics of participants, stratified by heart disease status.

Table 2. Baseline Characteristics of Study Participants by Heart Disease Status

Characteristic	0 N = 410 ¹	1 N = 508 ¹	p-value ²
Age	51 (9)	56 (9)	<0.001
Sex			<0.001
F	143 (35%)	50 (9.8%)	
M	267 (65%)	458 (90%)	
ChestPainType			<0.001
ASY	104 (25%)	392 (77%)	
ATA	149 (36%)	24 (4.7%)	
NAP	131 (32%)	72 (14%)	
TA	26 (6.3%)	20 (3.9%)	
RestingBP	130 (16)	134 (20)	<0.001

Participants who developed heart disease were, on average, older than those who did not (mean age 56 vs. 51 years), and the heart disease group was disproportionately male (90% vs. 65% in the disease-free group), consistent with well-established sex-based differences in cardiovascular risk. Asymptomatic chest pain (ASY) was markedly more common among participants with heart disease (77%) than among those without (25%), whereas atypical anginal (ATA) and non-anginal (NAP) presentations were more frequent in the disease-free group. Resting blood pressure was modestly higher among participants with heart disease. These patterns are consistent with the established epidemiology of coronary artery disease and provide an appropriate descriptive foundation for the bivariate and multivariable analyses that follow.

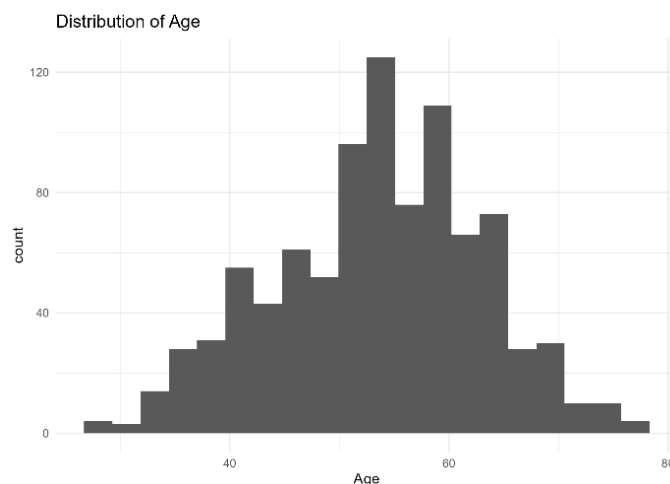
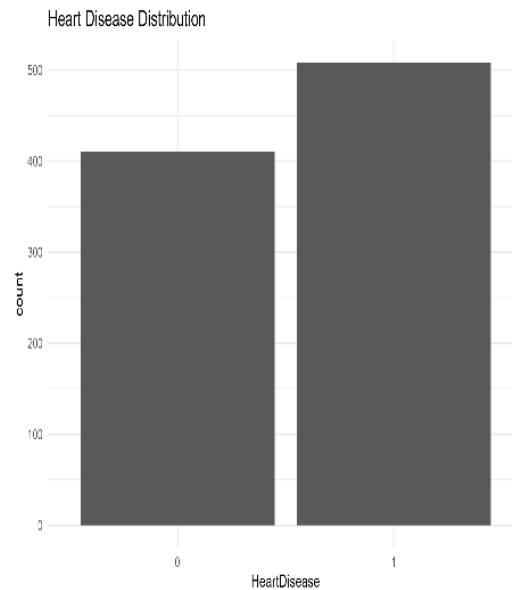


Figure 1. Distribution of Age Among Study Participants

Participant age ranged from approximately 28 to 77 years, with the distribution concentrated between 45 and 65 years and a peak in the early-to-mid 50s. The roughly symmetric shape of the distribution supports the use of age as a continuous predictor in the regression analysis.

**Figure 2. Distribution of Heart Disease Status Among Participants**

Of the 918 participants, 508 (55.3%) had heart disease and 410 (44.7%) did not, indicating a reasonably balanced outcome distribution. This near-even split is favorable for logistic regression modeling and underlies the no-information rate of 55.3% referenced later in the model performance evaluation (Table 6).

**Figure 3. Age Distribution by Heart Disease Status**

Participants with heart disease had a visibly higher median age (57 years) than participants without heart disease (51 years), with the interquartile range also shifted upward in the disease group. A small number of younger outliers within the heart disease group indicate that age alone does not fully account for disease risk, reinforcing the rationale for a multivariable analytic approach.

5.2 Correlation Among Continuous Predictors

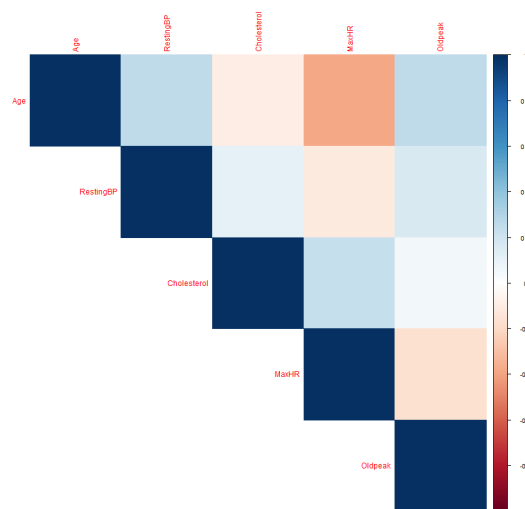


Figure 4. Correlation Heatmap of Continuous Predictor Variables

Pairwise correlations among the continuous predictors — age, resting blood pressure, cholesterol, maximum heart rate, and oldpeak — were generally weak to moderate, with no coefficient suggesting severe linear dependence. The strongest relationship observed was an inverse association between age and maximum heart rate, consistent with the well-documented physiological decline in peak heart rate with advancing age. The absence of strong inter-correlations supports the appropriateness of including these variables together in a single multivariable logistic regression model.

5.3 Bivariate Analysis

Bivariate associations between selected predictor variables and heart disease status were assessed using independent-samples Welch t-tests for continuous variables and Pearson chi-square tests for categorical variables (Table 3).

Table 3. Bivariate Associations Between Selected Variables and Heart Disease

Variable	Statistical Test	Test Statistic	Degrees of Freedom	P-value
Age	Welch Two-Sample t-test	$t = -8.82$	843.69	<0.001
Cholesterol	Welch Two-Sample t-test	$t = 7.63$	844.36	<0.001
Sex	Pearson Chi-square Test	$\chi^2 = 84.15$	1	<0.001
Exercise-Induced Angina	Pearson Chi-square Test	$\chi^2 = 222.26$	1	<0.001

Age and cholesterol differed significantly between participants with and without heart disease ($p < 0.001$ for both), with affected participants being older on average and showing lower mean cholesterol values. Sex and exercise-induced angina were also significantly associated with heart disease status ($p < 0.001$ for both), with male sex and the presence of exercise-induced angina markedly more common among participants with heart disease. These variables, together with the remaining candidate predictors, were carried forward into the multivariable logistic regression model.

5.4 Multivariable Logistic Regression Analysis

A multivariable logistic regression model was fitted to identify independent predictors of heart disease while adjusting for all other variables in the model. Adjusted odds ratios (AOR) with 95% confidence intervals are presented in Table 4.

Table 4. Multivariable Logistic Regression Analysis of Factors Associated with Heart Disease

Variable	AOR (95% CI)	P-value
Age	1.02 (0.99–1.04)	0.210
Sex (Male vs Female)	4.33 (2.52–7.57)	<0.001
Chest Pain Type: ATA vs ASY	0.16 (0.08–0.30)	<0.001
Chest Pain Type: NAP vs ASY	0.19 (0.11–0.31)	<0.001
Chest Pain Type: TA vs ASY	0.23 (0.10–0.52)	<0.001
Resting Blood Pressure	1.00 (0.99–1.02)	0.485
Cholesterol	0.996 (0.994–0.998)	<0.001
Fasting Blood Sugar (1 vs 0)	3.12 (1.83–5.40)	<0.001
Resting ECG: Normal vs LVH	0.84 (0.49–1.43)	0.515
Resting ECG: ST vs LVH	0.76 (0.38–1.52)	0.443
Maximum Heart Rate	1.00 (0.99–1.01)	0.393
Exercise-Induced Angina (Yes vs No)	2.46 (1.52–3.98)	<0.001
Oldpeak	1.46 (1.16–1.85)	0.001
ST Slope: Flat vs Down	4.28 (1.81–9.79)	<0.001
ST Slope: Up vs Down	0.37 (0.15–0.88)	0.027

After adjustment, several variables emerged as independent predictors of heart disease. Male sex was associated with more than a four-fold increase in the odds of heart disease relative to female sex (AOR 4.33, 95% CI 2.52–7.57, $p<0.001$). Relative to asymptomatic chest pain (ASY), all other chest pain types were associated with substantially lower odds of heart disease, suggesting that an asymptomatic presentation is itself a strong marker of underlying disease in this cohort. Fasting blood sugar above 120 mg/dl (AOR 3.12, $p<0.001$), exercise-induced angina (AOR 2.46, $p<0.001$), a flat ST slope relative to a downsloping segment (AOR 4.28, $p<0.001$), and greater oldpeak (ST depression; AOR 1.46 per unit, $p=0.001$) were each associated with significantly higher odds of heart disease, consistent with their established roles as clinical markers of myocardial ischemia. Resting blood pressure, maximum heart rate, and resting ECG findings were not independently associated with heart disease after adjustment.

Cholesterol showed a small but statistically significant inverse association with heart disease in the adjusted model (AOR 0.996, $p<0.001$). Because this direction of effect runs counter to established clinical evidence linking elevated cholesterol to cardiovascular risk, it most likely reflects known data-quality artifacts in this public dataset — including biologically implausible recorded values — rather than a genuine protective effect, and should be interpreted with caution rather than as a clinical finding.

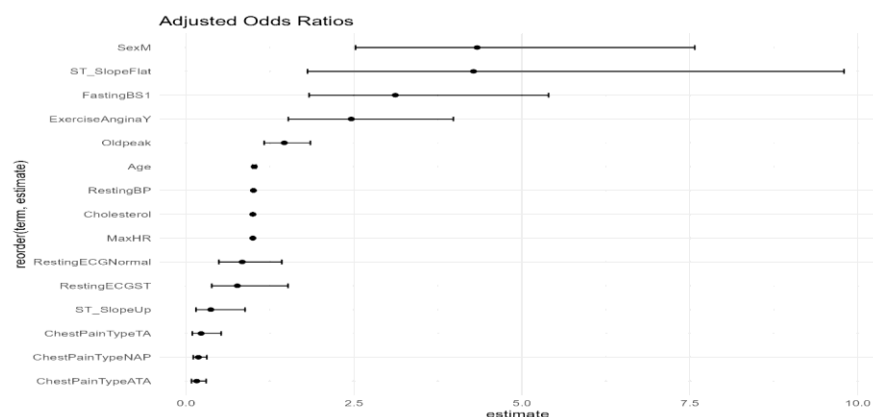


Figure 5. Forest Plot of Adjusted Odds Ratios for Predictors of Heart Disease

The forest plot summarizes the adjusted odds ratios and 95% confidence intervals from the multivariable model. Male sex and a flat ST slope show the largest point estimates but also the widest confidence intervals, reflecting greater uncertainty around their precise magnitude despite both being highly statistically significant. Variables whose confidence intervals cross 1.0 — including resting blood pressure, maximum heart rate, and resting ECG categories — were not statistically significant predictors in the adjusted model.

5.5 Multicollinearity Diagnostics

Multicollinearity among predictor variables was assessed using generalized variance inflation factors (GVIF), adjusted for the degrees of freedom of each categorical variable (Table 5).

Table 5. Assessment of Multicollinearity Among Predictor Variables

Variable	GVIF	Df	Adjusted GVIF*
Age	1.330	1	1.153
Sex	1.101	1	1.049
Chest Pain Type	1.229	3	1.035
Resting Blood Pressure	1.133	1	1.065
Cholesterol	1.227	1	1.108
Fasting Blood Sugar	1.088	1	1.043
Resting ECG	1.237	2	1.055
Maximum Heart Rate	1.365	1	1.168
Exercise-Induced Angina	1.221	1	1.105
Oldpeak	1.329	1	1.153
ST Slope	1.496	2	1.106

*Adjusted GVIF = $GVIF^{1/(2 \times Df)}$

All adjusted GVIF values were well below the conventional threshold of 2, with the highest value observed for ST slope (1.106). This indicates no evidence of problematic multicollinearity among the predictor variables, supporting the stability and reliable interpretation of the regression coefficients reported in Table 4.

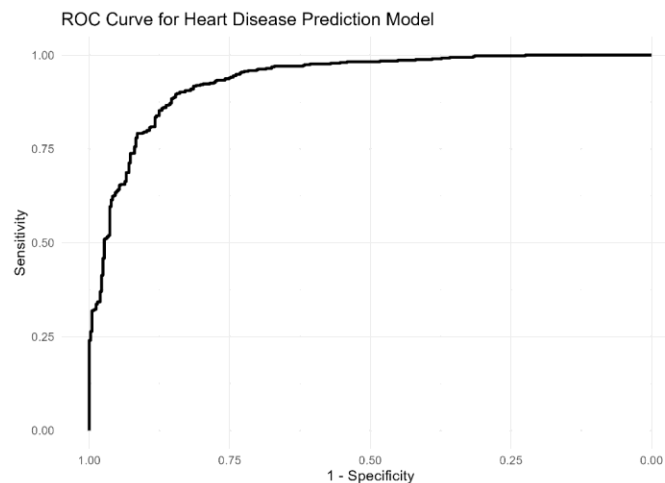
5.6 Model Performance and Discrimination

The predictive performance of the final logistic regression model was evaluated using standard classification metrics (Table 6).

Table 6. Performance of the Logistic Regression Model for Predicting Heart Disease

Performance Metric	Value
Accuracy	87.0%
Sensitivity	83.2%
Specificity	90.2%
Positive Predictive Value (PPV)	87.2%
Negative Predictive Value (NPV)	86.9%
Balanced Accuracy	86.7%
Kappa Statistic	0.737
95% Confidence Interval for Accuracy	84.7% – 89.1%
No Information Rate (NIR)	55.3%
P-value (Accuracy > NIR)	<0.001

The final model correctly classified 87.0% of participants, with a sensitivity of 83.2% (the proportion of true heart disease cases correctly identified) and a specificity of 90.2% (the proportion of disease-free participants correctly identified). The kappa statistic of 0.737 indicates substantial agreement between predicted and observed classifications beyond chance. Model accuracy was significantly higher than the no-information rate of 55.3% ($p < 0.001$), confirming that the model provides meaningful predictive value beyond simply predicting the majority class.

**Figure 6. ROC Curve for the Heart Disease Prediction Model****Table 7. Area Under the Receiver Operating Characteristic (ROC) Curve**

Performance Measure	Value
Area Under the Curve (AUC)	0.933

The ROC curve illustrates the trade-off between sensitivity and specificity across all possible classification thresholds, bowing sharply toward the top-left corner — a visual signature of strong discriminatory performance. This is confirmed by an area under the curve (AUC) of 0.933, which falls within the “excellent” range for model discrimination and indicates that the model can correctly distinguish a randomly selected participant with heart disease from one without it in 93.3% of such comparisons.

Chapter 6: Public Health Implications

The findings of this analysis carry several implications for cardiovascular disease prevention and health system planning. The independent predictors identified — sex, chest pain presentation, fasting blood sugar, exercise-induced angina, oldpeak, and ST slope — are all variables that can be captured through routine clinical history, basic laboratory testing, and a standard exercise electrocardiogram. This makes the underlying model attractive for use as a low-cost, low-technology risk-stratification tool in primary care and outpatient settings, particularly in resource-constrained environments where advanced cardiac imaging is not readily accessible.

The strong discriminatory performance of the model (AUC = 0.933, accuracy = 87.0%) suggests that simple, interpretable statistical tools can meaningfully support clinical decision-making without requiring complex or expensive diagnostic infrastructure. Embedding a model of this kind into routine screening protocols at the primary healthcare level could help frontline clinicians flag high-risk individuals for referral, closer monitoring, or early lifestyle and pharmacological intervention, rather than relying solely on clinical judgment or symptom presentation alone.

The markedly elevated risk associated with fasting blood sugar above 120 mg/dl and with exercise-induced angina reinforces the importance of integrated non-communicable disease (NCD) programs that screen for diabetes and cardiovascular risk jointly rather than in isolation. Given the well-documented overlap between metabolic and cardiovascular risk factors, public health programs that combine glycemic screening with cardiovascular risk assessment are likely to identify high-risk individuals more efficiently than single-disease screening pathways.

From a health systems perspective, the pronounced sex disparity observed in this cohort (male sex associated with over four times the odds of heart disease) underscores the need for targeted awareness and screening campaigns aimed at men, who may be less likely to seek preventive care, while also ensuring that cardiovascular risk in women is not under-recognized due to differing symptom presentation. Finally, this project demonstrates the practical value of applying accessible biostatistical methods — descriptive analysis, bivariate testing, and multivariable logistic regression — to generate actionable, locally interpretable evidence, illustrating a replicable analytic workflow that can be adapted to other NCD risk-prediction problems in similar public health settings, including low- and middle-income countries such as Ethiopia.

Chapter 7: Limitations

Several limitations should be considered when interpreting the findings of this analysis. First, the cross-sectional design of the dataset means that predictor and outcome variables were measured at a single point in time; this precludes any causal interpretation of the associations identified, which should be understood as statistical associations rather than evidence of cause and effect.

Second, the dataset was obtained from a publicly available source on Kaggle and represents a secondary compilation of clinical records from multiple original studies. The provenance, measurement protocols, and data collection procedures of the original sources are not fully documented, and some apparent data-quality artifacts were observed — most notably the counter-intuitive inverse association between cholesterol and heart disease, which likely reflects implausible recorded values (e.g., zero entries) rather than a genuine biological relationship. This limits confidence in the cholesterol-related findings specifically.

Third, the study population was drawn from individuals undergoing clinical cardiovascular assessment and is therefore unlikely to represent the general adult population; the prevalence of heart disease and the distribution of risk factors observed here may not generalize to community-based or population-level samples, particularly in different geographic or healthcare-system contexts such as Ethiopia.

Fourth, the dataset lacks several variables known to be important confounders or predictors of cardiovascular disease, including smoking status, physical activity level, dietary habits, family history of cardiovascular disease, body mass index, and socioeconomic indicators. The absence of these variables means that the model may be subject to residual confounding, and the true strength of some associations may be over- or under-estimated.

Finally, the model was developed and evaluated on the same dataset without an independent external validation cohort. While internal performance metrics (accuracy, sensitivity, specificity, AUC) were strong, these estimates may be optimistic relative to performance on new, unseen populations. Prospective validation in an independent cohort, ideally drawn from the intended target population, would be required before any clinical or programmatic application of this model.

Chapter 8: Conclusion

This project applied descriptive, bivariate, and multivariable logistic regression methods to identify predictors of heart disease among adults using a publicly available clinical dataset. Male sex, asymptomatic chest pain presentation, elevated fasting blood sugar, exercise-induced angina, greater ST depression (oldpeak), and a flat ST slope were each identified as independent and clinically plausible predictors of heart disease, consistent with established cardiovascular pathophysiology.

The final model demonstrated strong predictive performance, correctly classifying 87.0% of participants with an area under the ROC curve of 0.933, indicating excellent discriminatory ability between individuals with and without heart disease. Diagnostic checks confirmed the absence of problematic multicollinearity among predictors, supporting the reliability of the model's coefficient estimates.

Beyond the specific findings, this project illustrates the broader value of biostatistical methods in public health research: transforming routinely collected clinical variables into an interpretable, evidence-based tool for identifying individuals at elevated cardiovascular risk. Such tools have practical relevance for primary care screening and non-communicable disease prevention programs, particularly in settings where advanced diagnostic resources are limited.

Future work should prioritize external validation of the model in independent and, ideally, population-representative cohorts; incorporation of additional risk factors such as smoking, physical activity, body mass index, and family history; and exploration of alternative and complementary modeling approaches — including machine learning methods — to benchmark predictive performance against the logistic regression framework applied here. Prospective, longitudinal data collection would also allow for a more rigorous assessment of the causal pathways underlying the associations identified in this report.